

Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
plt.style.use('seaborn-v0_8-pastel')

import seaborn as sns

%matplotlib inline
# used to display your plots within the notebook.
```

Load and Explore the Dataset

```
# Importing the dataset
df = pd.read_csv(r"C:\Users\yasha\OneDrive\Desktop\inlighthntech.com\My projects\Python\
New Year Sales Data.csv", encoding='latin-1')
```

```
# Checking the dataset
df.head()
```

	User_ID	Cust_name	Product_ID	Gender	Age	Group	Age	Marital_Status	\
0	1002903	Sanskriti	P00125942	F	26-35	28		0	
1	1000732	Kartik	P00110942	F	26-35	35		1	
2	1001990	Bindu	P00118542	F	26-35	35		1	
3	1001425	Sudevi	P00237842	M	0-17	16		0	
4	1000588	Joni	P00057942	M	26-35	28		1	

	State	Zone	Occupation	Product_Category	Orders	\
0	Maharashtra	Western	Healthcare	Auto	1	
1	Andhra Pradesh	Southern	Govt	Auto	3	
2	Uttar Pradesh	Central	Automobile	Auto	3	
3	Karnataka	Southern	Construction	Auto	2	
4	Gujarat	Western	Food Processing	Auto	2	

	Amount	Status	unnamed1
0	23952.0	NaN	NaN
1	23934.0	NaN	NaN
2	23924.0	NaN	NaN
3	23912.0	NaN	NaN
4	23877.0	NaN	NaN

```
# Checking the rows and columns
df.shape
```

```
(11251, 15)
```

```
# Checking info about column names, non-null values and Data types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype
---  -
#   Column              Non-Null Count  Dtype
```

0	User_ID	11251	non-null	int64
1	Cust_name	11251	non-null	object
2	Product_ID	11251	non-null	object
3	Gender	11251	non-null	object
4	Age Group	11251	non-null	object
5	Age	11251	non-null	int64
6	Marital_Status	11251	non-null	int64
7	State	11251	non-null	object
8	Zone	11251	non-null	object
9	Occupation	11251	non-null	object
10	Product_Category	11251	non-null	object
11	Orders	11251	non-null	int64
12	Amount	11239	non-null	float64
13	Status	0	non-null	float64
14	unnamed1	0	non-null	float64

dtypes: float64(3), int64(4), object(8)

memory usage: 1.3+ MB

Data Cleaning

```
# Checking null values
```

```
df.isnull().sum()
```

User_ID	0
Cust_name	0
Product_ID	0
Gender	0
Age Group	0
Age	0
Marital_Status	0
State	0
Zone	0
Occupation	0
Product_Category	0
Orders	0
Amount	12
Status	11251
unnamed1	11251

dtype: int64

```
# Dropping the null values and unnecessary columns
```

```
df.drop(["Status", "unnamed1"], axis= 1, inplace=True, errors='ignore')
```

```
# Dropping rows with null values in other important columns.
```

```
important_cols = ['Gender', 'Age Group', 'Marital_Status', 'State', 'Zone',  
'Occupation', 'Product_Category', 'Orders', 'Amount']
```

```
df = df.dropna(subset=important_cols)
```

```
df.isnull().sum()
```

User_ID	0
Cust_name	0
Product_ID	0
Gender	0
Age Group	0
Age	0

```
Marital_Status      0
State                0
Zone                0
Occupation           0
Product_Category    0
Orders              0
Amount              0
dtype: int64
```

```
#Converting the Amount column to an integer type
df['Amount'] = df['Amount'].astype(int)
```

Data Overview and Summary

```
# Statistical Summary
df.describe()
```

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

```
# Checking for unique values
df.nunique()
```

User_ID	3752
Cust_name	1250
Product_ID	2350
Gender	2
Age Group	7
Age	81
Marital_Status	2
State	16
Zone	5
Occupation	15
Product_Category	18
Orders	4
Amount	6583
dtype:	int64

```
for col in ['Gender', 'Age Group', 'Marital_Status', 'State', 'Zone', 'Occupation',
'Product_Category']:
    if col in df.columns:
        print(f"\nUnique values in {col}:")
        print(df[col].unique())
```

Unique values in Gender:
['F' 'M']

Unique values in Age Group:
['26-35' '0-17' '18-25' '51-55' '46-50' '55+' '36-45']

```
Unique values in Marital_Status:  
[0 1]
```

```
Unique values in State:  
['Maharashtra' 'Andhra\xa0Pradesh' 'Uttar Pradesh' 'Karnataka' 'Gujarat'  
 'Himachal Pradesh' 'Delhi' 'Jharkhand' 'Kerala' 'Haryana'  
 'Madhya Pradesh' 'Bihar' 'Rajasthan' 'Uttarakhand' 'Telangana' 'Punjab']
```

```
Unique values in Zone:  
['Western' 'Southern' 'Central' 'Northern' 'Eastern']
```

```
Unique values in Occupation:  
['Healthcare' 'Govt' 'Automobile' 'Construction' 'Food Processing'  
 'Lawyer' 'Media' 'Banking' 'Retail' 'IT Sector' 'Aviation' 'Hospitality'  
 'Agriculture' 'Textile' 'Chemical']
```

```
Unique values in Product_Category:  
['Auto' 'Hand & Power Tools' 'Stationery' 'Tupperware' 'Footwear & Shoes'  
 'Furniture' 'Food' 'Games & Toys' 'Sports Products' 'Books'  
 'Electronics & Gadgets' 'Decor' 'Clothing & Apparel' 'Beauty'  
 'Household items' 'Pet Care' 'Veterinary' 'Office']
```

```
df = df.apply(lambda col: col.str.replace('\xa0', ' ', regex=False) if col.dtype ==  
'object' else col)
```

```
print(df['State'].unique())
```

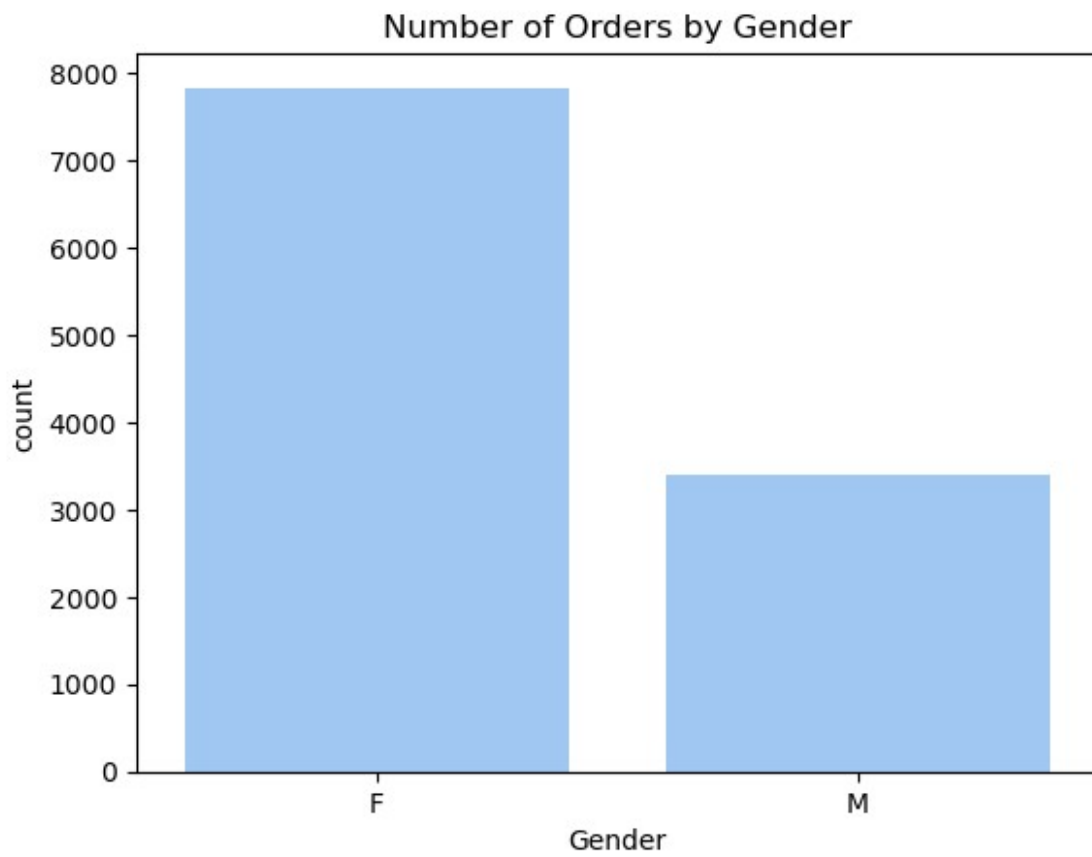
```
['Maharashtra' 'Andhra Pradesh' 'Uttar Pradesh' 'Karnataka' 'Gujarat'  
 'Himachal Pradesh' 'Delhi' 'Jharkhand' 'Kerala' 'Haryana'  
 'Madhya Pradesh' 'Bihar' 'Rajasthan' 'Uttarakhand' 'Telangana' 'Punjab']
```

Exploratory Data Analysis (EDA)

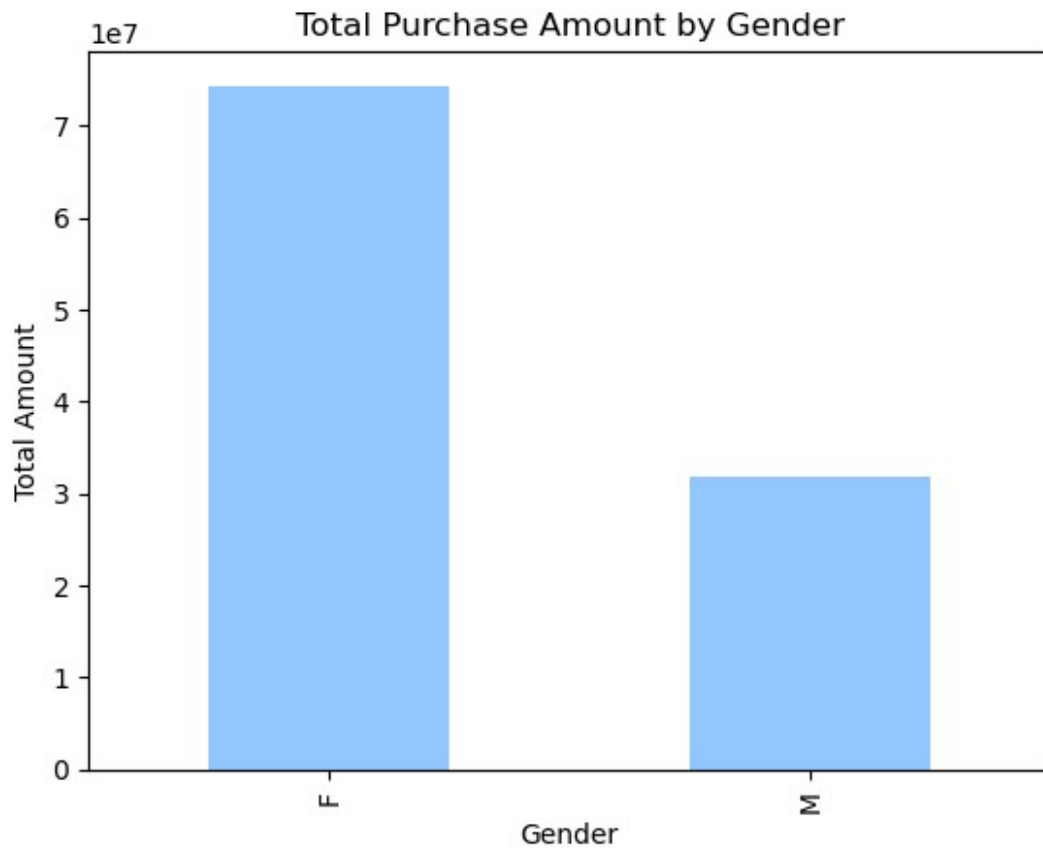
Gender Analysis

Which gender has a higher purchasing power?

```
# Count Plot  
sns.countplot(data=df, x='Gender')  
plt.title("Number of Orders by Gender")  
plt.show()
```



```
gender_amount = df.groupby('Gender')['Amount'].sum()
gender_amount.plot(kind='bar')
plt.title("Total Purchase Amount by Gender")
plt.ylabel("Total Amount")
plt.show()
```



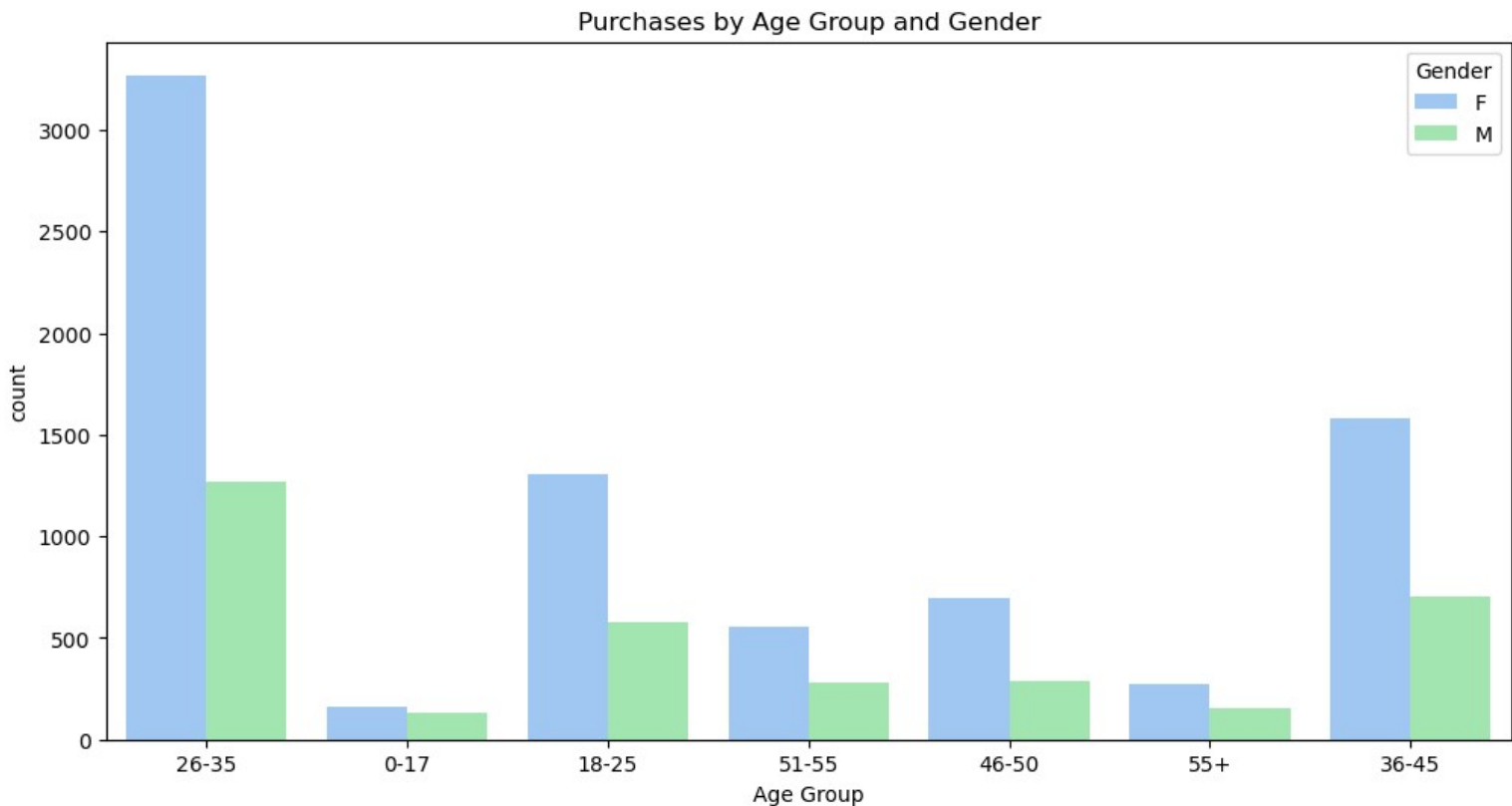
Findings Summary

- Female (F) customers generated approximately ₹7.4 crore (~74M) in total purchases
- Male (M) customers generated approximately ₹3.1 crore (~31M) in total purchases
- Females outspend males by roughly x2.4, making gender a strong differentiator in revenue contribution ~70% of total sales revenue
- Marketing and inventory decisions should prioritize female customer preferences
- Males, while likely present in reasonable order volume, have a notably lower spend per segment — worth investigating whether this is due to fewer orders or lower average order value

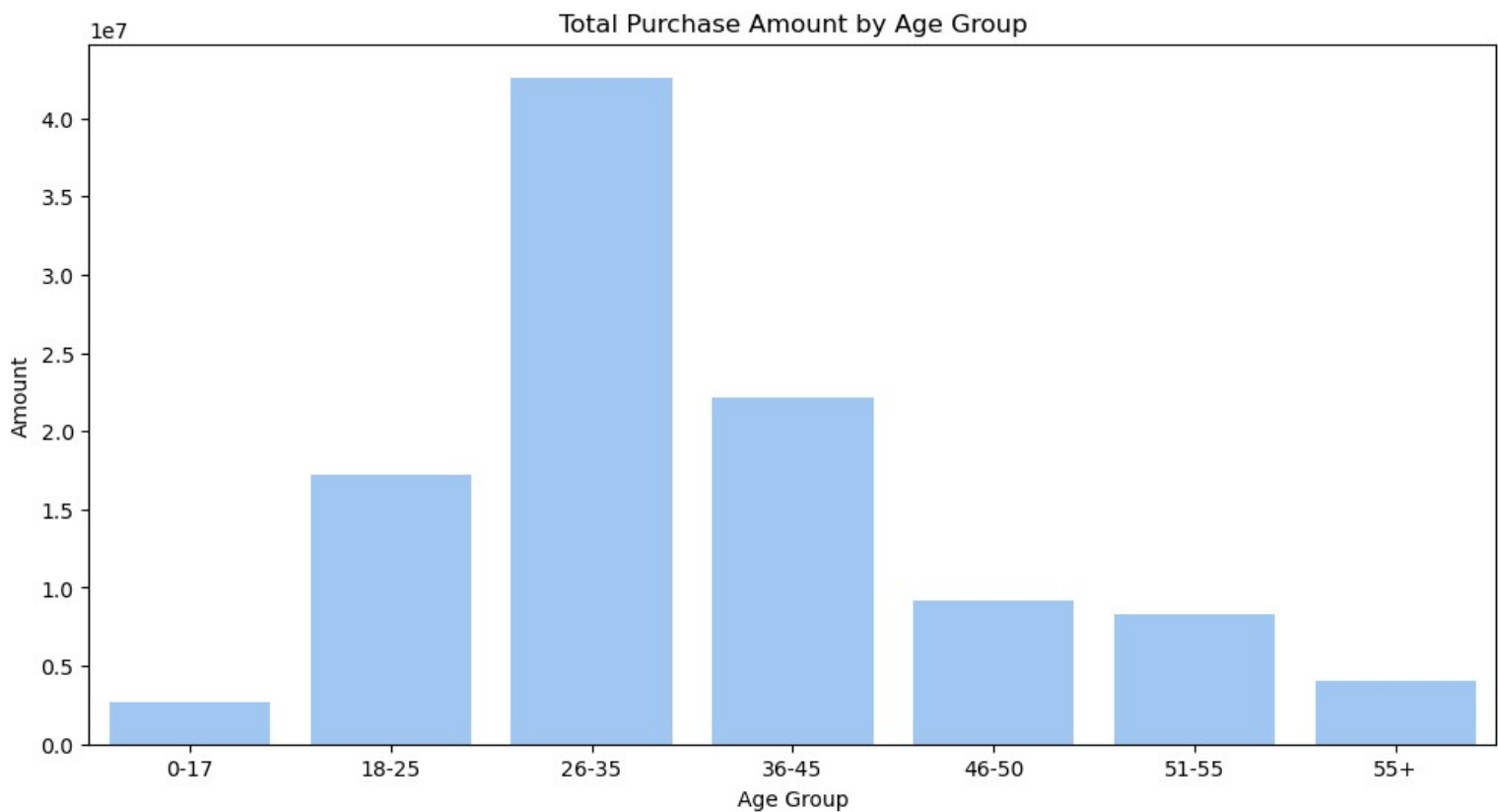
Age Group Analysis

Which age group has the most purchases, and is there a trend in purchasing power by age?

```
# count plot of age groups with hue as gender
plt.figure(figsize=(12,6))
sns.countplot(data=df, x='Age Group', hue="Gender")
plt.title("Purchases by Age Group and Gender")
plt.show()
```



```
# bar chart showing total purchase amount by age group
age_amount = df.groupby('Age Group')['Amount'].sum().reset_index()
plt.figure(figsize=(12,6))
sns.barplot(data=age_amount, x='Age Group', y='Amount')
plt.title("Total Purchase Amount by Age Group")
plt.show()
```



Findings Summary

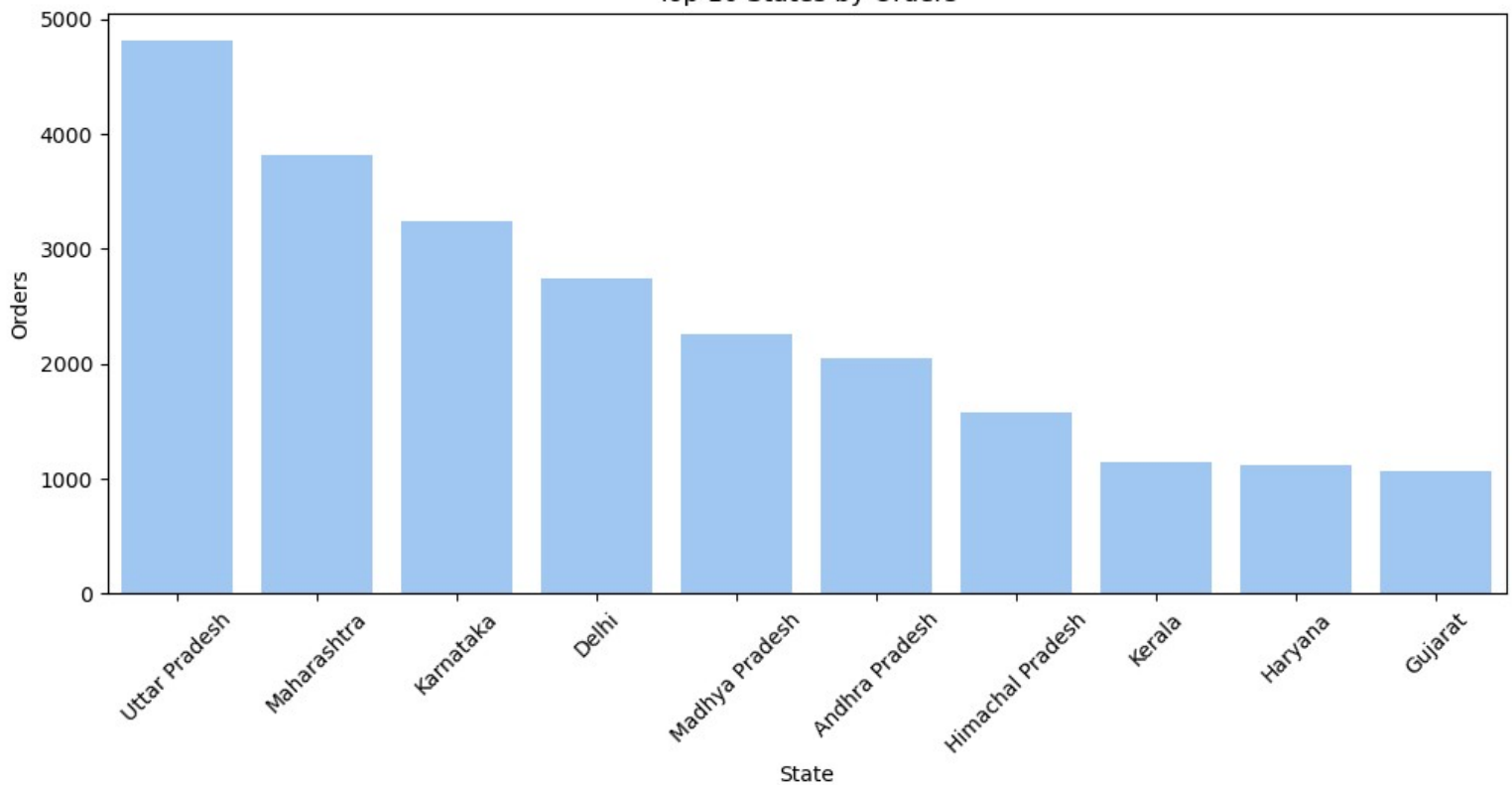
- The 26–35 age group dominates, contributing nearly 40% of total purchase revenue — almost double the next closest group
- Spending drops sharply after 35, showing a clear inverted-V pattern peaking in the late 20s to mid-30s
- The 18–25 group shows decent spend, indicating younger adults are an emerging segment
- Customers 45 and older collectively contribute less than the 26–35 group alone

State Analysis

Which states generate the highest number of orders and revenue?

```
# bar charts for the number of orders and total amount by state
top_states_orders = df.groupby('State')['Orders'].sum().nlargest(10).reset_index()
plt.figure(figsize=(12,5))
sns.barplot(data=top_states_orders, x='State', y='Orders')
plt.xticks(rotation=45)
plt.title("Top 10 States by Orders")
plt.show()
```

Top 10 States by Orders

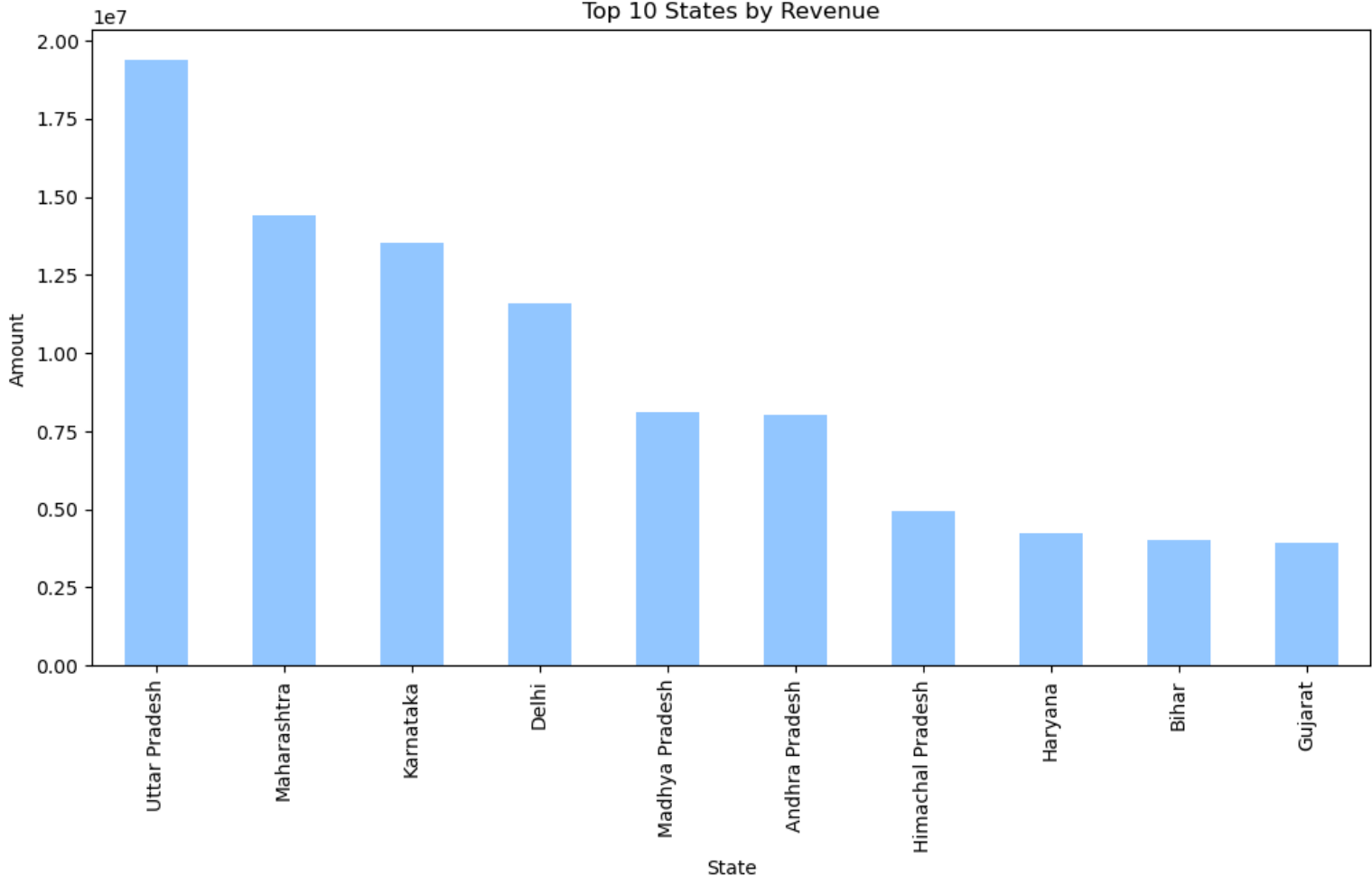


Key Observations

- Uttar Pradesh leads by a wide margin, generating ~26% more orders than Maharashtra
- The top 3 states (UP, Maharashtra, Karnataka) together account for the bulk of all orders
- There's a steep drop-off after Delhi — the bottom 4 states each have fewer than 1,200 orders
- The high-order states are a mix of large population centers (UP, Maharashtra, Delhi) and tech/metro hubs (Karnataka)

```
plt.figure(figsize=(12,6))
top_state_amount = df.groupby('State')['Amount'].sum().nlargest(10)
top_state_amount.plot(kind='bar')
plt.title("Top 10 States by Revenue")
plt.ylabel('Amount')
plt.show()
```


Top 10 States by Revenue



Findings Summary

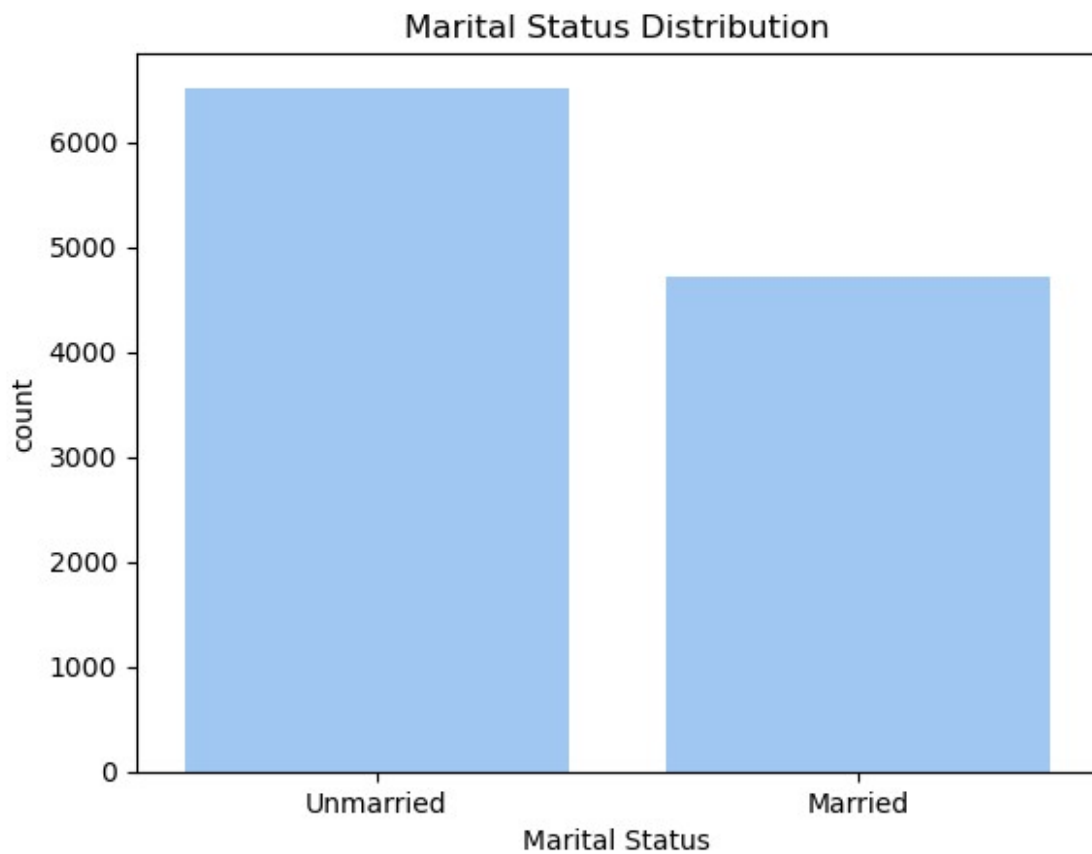
- Uttar Pradesh dominates revenue too, consistent with its #1 ranking in orders — high volume is clearly translating to high revenue
- The top 4 states (UP, Maharashtra, Karnataka, Delhi) are identical across both orders and revenue charts, confirming these are the core markets
- Bihar replaces Kerala from the orders chart, suggesting Bihar customers spend more per order despite placing fewer orders overall
- Revenue drops sharply after Delhi — the bottom 6 states each generate less than ₹0.85 crore

Marital Status Analysis

How does marital status affect purchasing behavior?

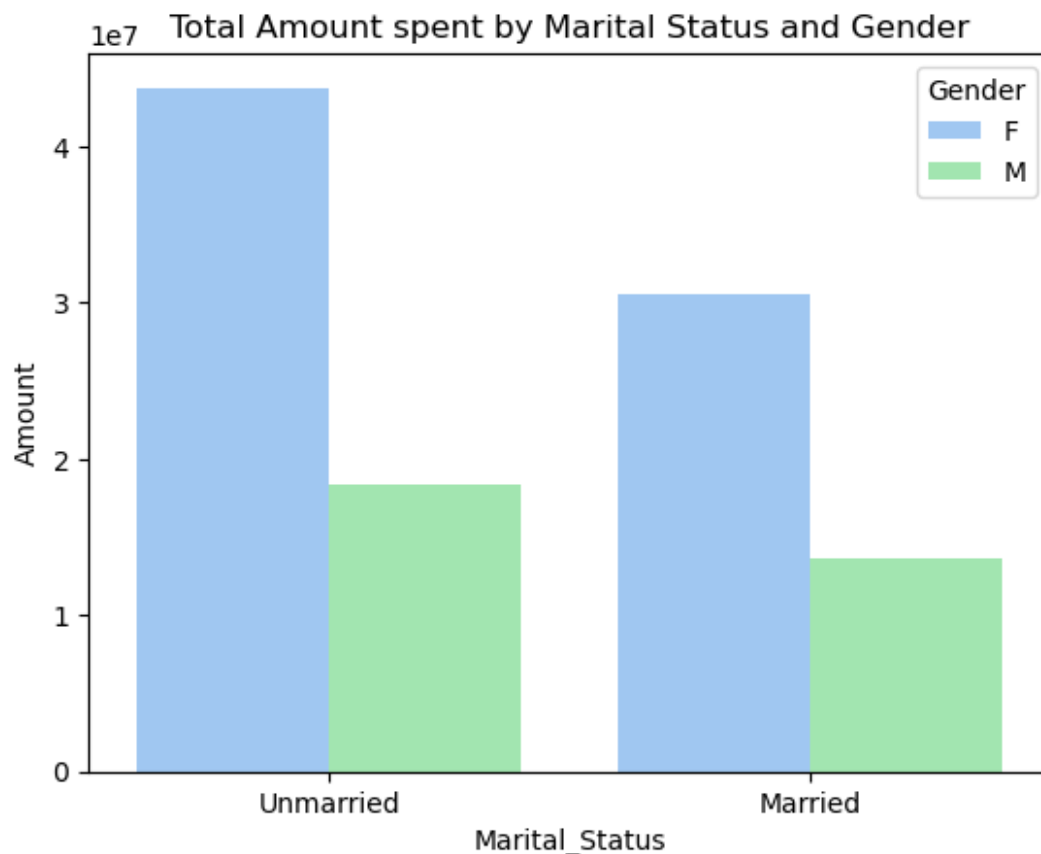
```
# count plot for marital status
ax = sns.countplot(data=df, x='Marital_Status')
plt.title('Marital Status Distribution')
ax.set_xticks([0,1])
ax.set_xticklabels(['Unmarried', 'Married'])
ax.set_xlabel('Marital Status')

plt.show()
```



bar chart of total amount spent by marital status, with gender as a hue

```
marital_amount = df.groupby(['Marital_Status', 'Gender'])  
['Amount'].sum().reset_index()  
  
ax = sns.barplot(data=marital_amount, x='Marital_Status', y='Amount', hue='Gender')  
plt.title("Total Amount spent by Marital Status and Gender")  
ax.set_xticks([0,1])  
ax.set_xticklabels(['Unmarried', 'Married'])  
plt.show()
```



Finding Summary

How Marital Status Affects Purchasing Behavior?

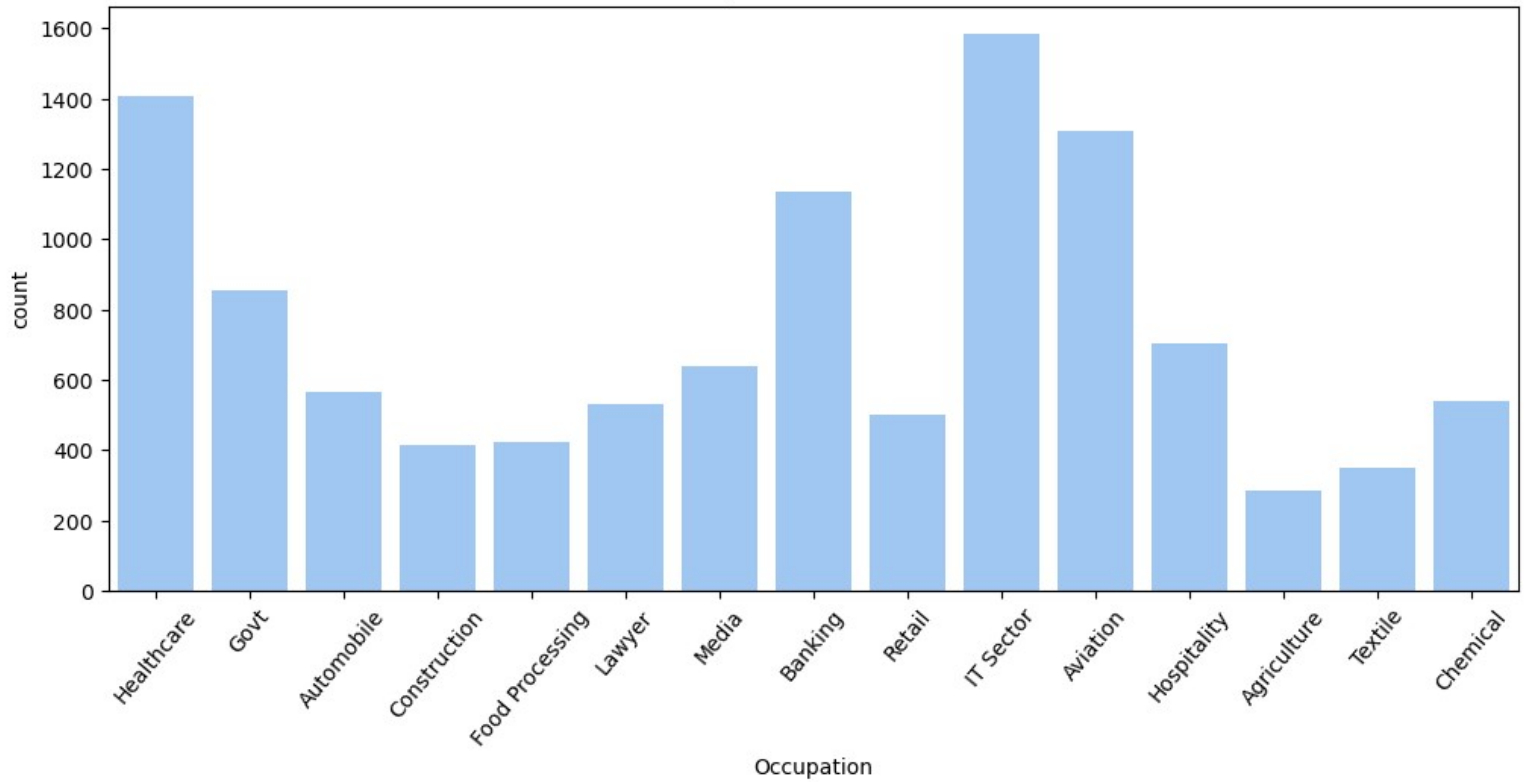
- Unmarried customers outspend married customers across both genders — unmarried females spend ~43% more than married females, and unmarried males spend ~37% more than married males
- Females dominate spending in both marital categories — unmarried females alone account for the largest single segment
- The pattern suggests unmarried individuals have higher discretionary spending, likely due to fewer shared financial obligations
- Across all four segments, the gender gap is consistent — females always outspend males regardless of marital status

Occupation Analysis

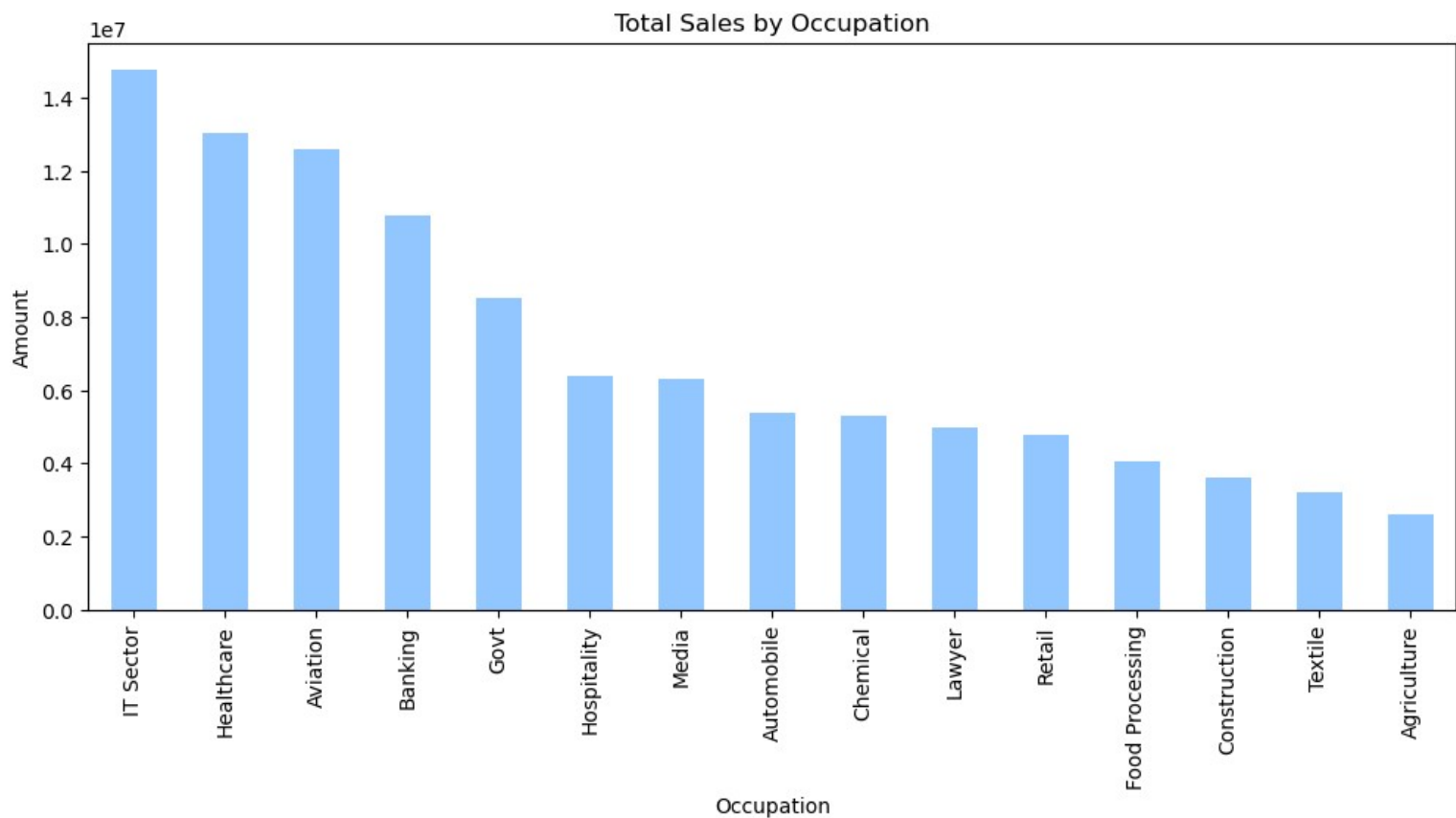
Which occupations contribute most to sales?

```
# count plot of occupation
plt.figure(figsize=(12,5))
sns.countplot(data=df, x='Occupation')
plt.title("Occupation Distribution")
plt.xticks(rotation=50)
plt.show()
```

Occupation Distribution



```
# barplot of total sales by occupation
plt.figure(figsize=(12,5))
occupation_amount = df.groupby('Occupation')
['Amount'].sum().sort_values(ascending=False)
occupation_amount.plot(kind='bar')
plt.title("Total Sales by Occupation")
plt.ylabel('Amount')
plt.show()
```



Findings Summary

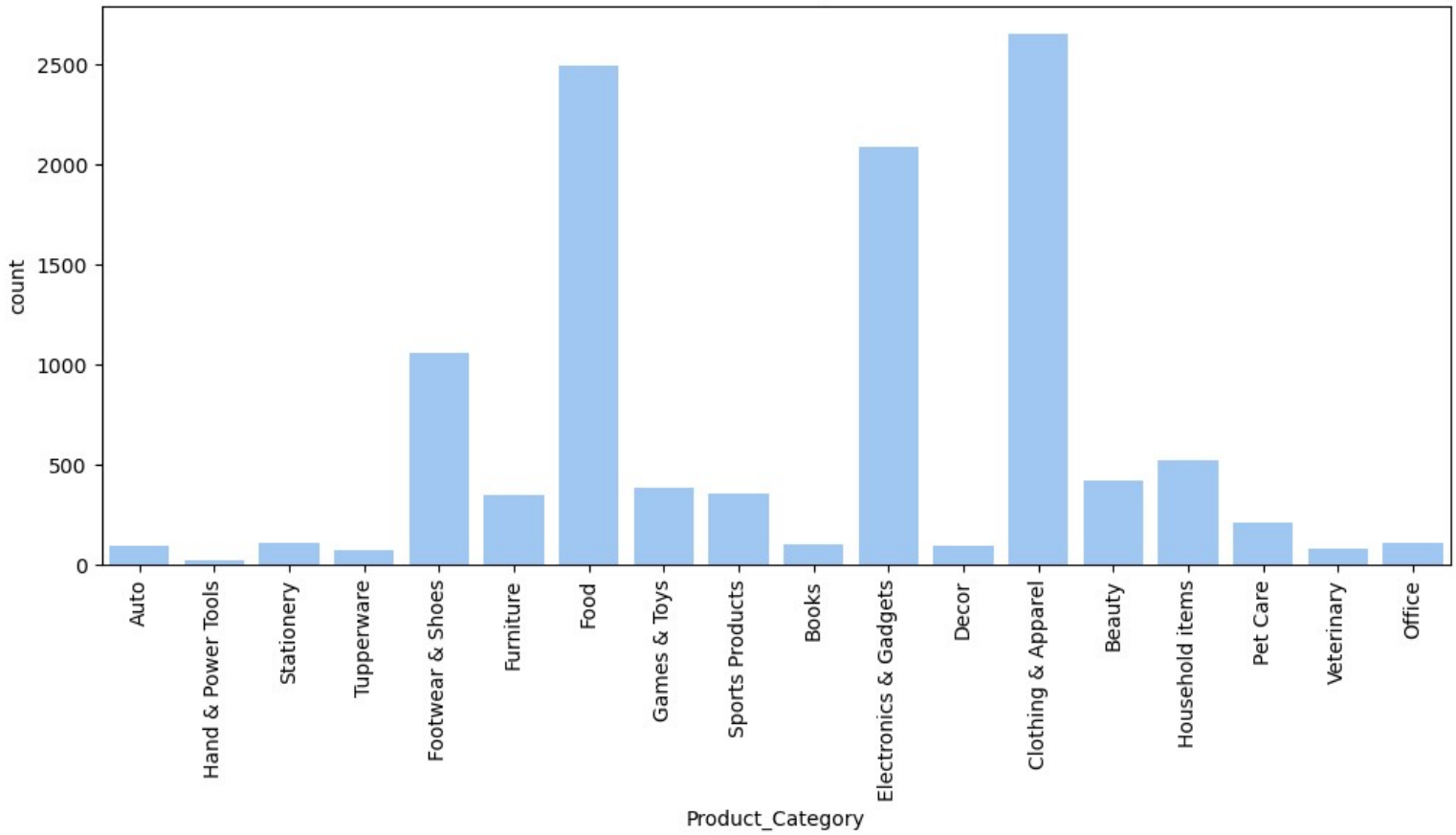
- IT Sector leads all occupations, consistent with high salaries and strong online shopping habits
- The top 4 (IT, Healthcare, Aviation, Banking) are all high-income professional sectors, collectively contributing the majority of sales revenue
- There's a sharp cliff after Banking — Govt and below contribute significantly less
- Agriculture contributes the least, reflecting lower disposable income and limited digital commerce penetration

Product Category Analysis

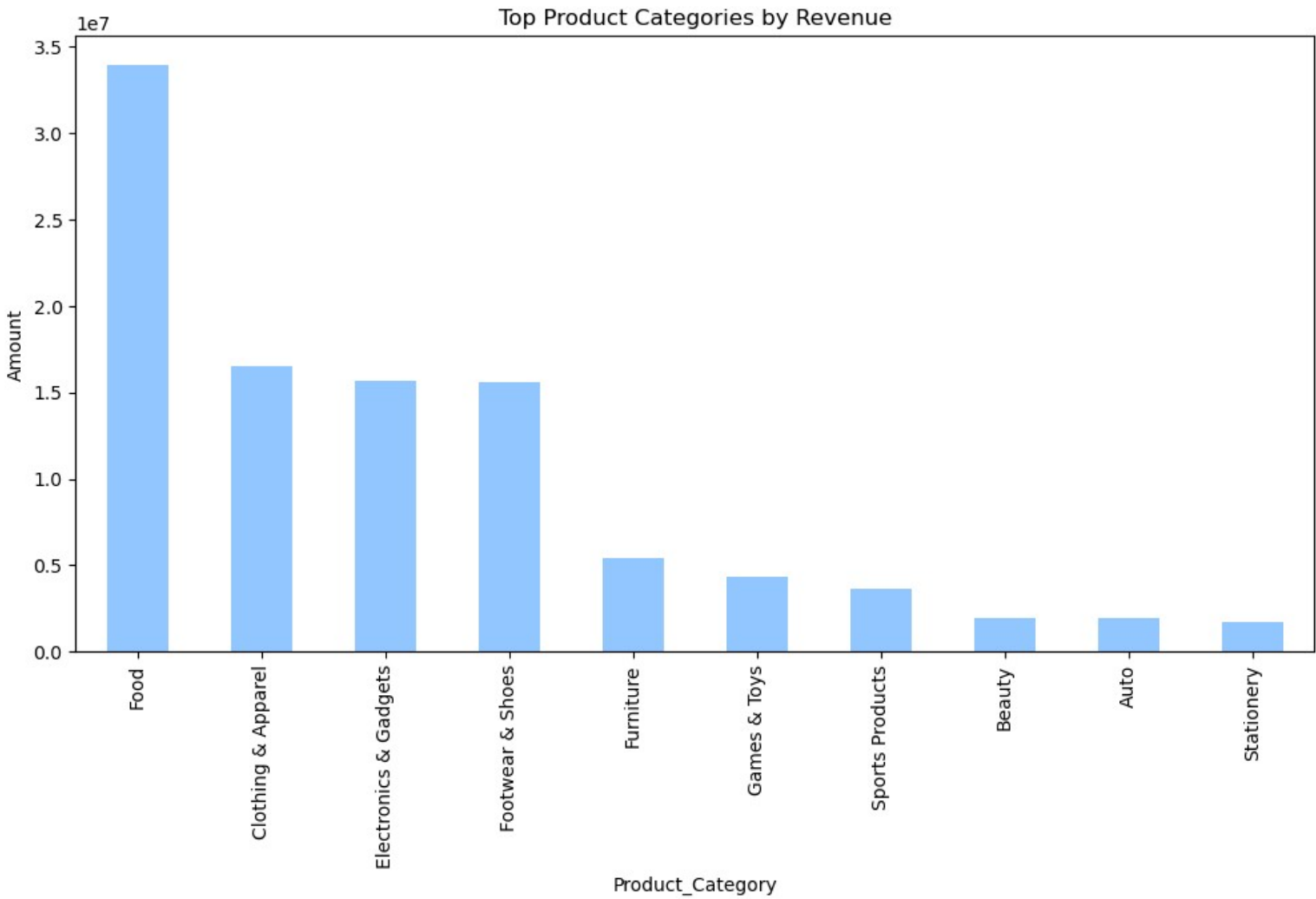
What product categories are the most popular, and which ones generate the most revenue?

```
# count plot of product categories
plt.figure(figsize=(12,5))
sns.countplot(data=df, x='Product_Category')
plt.title('Product Categories')
plt.xticks(rotation=90)
plt.show()
```

Product Categories



```
# bar chart of total amount by product category for the top categories
category_amount = df.groupby('Product_Category')['Amount'].sum().nlargest(10)
plt.figure(figsize=(12,6))
category_amount.plot(kind='bar')
plt.title("Top Product Categories by Revenue")
plt.ylabel('Amount')
plt.show()
```



Findings Summary

- Food dominates by a massive margin — generating ~2× the revenue of the next closest category, making it the single most critical product segment
- Clothing, Electronics, and Footwear form a strong second tier, each contributing ~₹1.55 crore and nearly neck-and-neck with each other
- There's a dramatic cliff after Footwear — Furniture onwards contributes less than ₹0.55 crore each
- Beauty, Auto, and Stationery are the weakest categories, each under ₹0.20 crore
- The top 4 categories together account for roughly 80% of total revenue

Conclusion

Complete High-Purchasing Customer Profile

Attribute	Profile
Gender	Female
Marital Status	Unmarried
Age Group	26–35
Occupation	IT
Sector	Healthcare, or Aviation
Location	Uttar Pradesh, Maharashtra, Karnataka, or Delhi
Top Categories	Food, Clothing & Apparel, Electronics & Gadgets, Footwear
Life Stage	Independent, salaried young professional

Attribute	Profile
Spending Driver	High disposable income; everyday essentials + lifestyle upgrades

Final Business Insight:

- Stock and promote Food, Clothing, Electronics, and Footwear heavily during New Year sales
- These four categories alone drive ~80% of revenue
- For the core segment — unmarried females aged 26–35 in IT/Healthcare — bundled offers across Clothing + Footwear + Electronics would likely yield the highest conversion and basket size

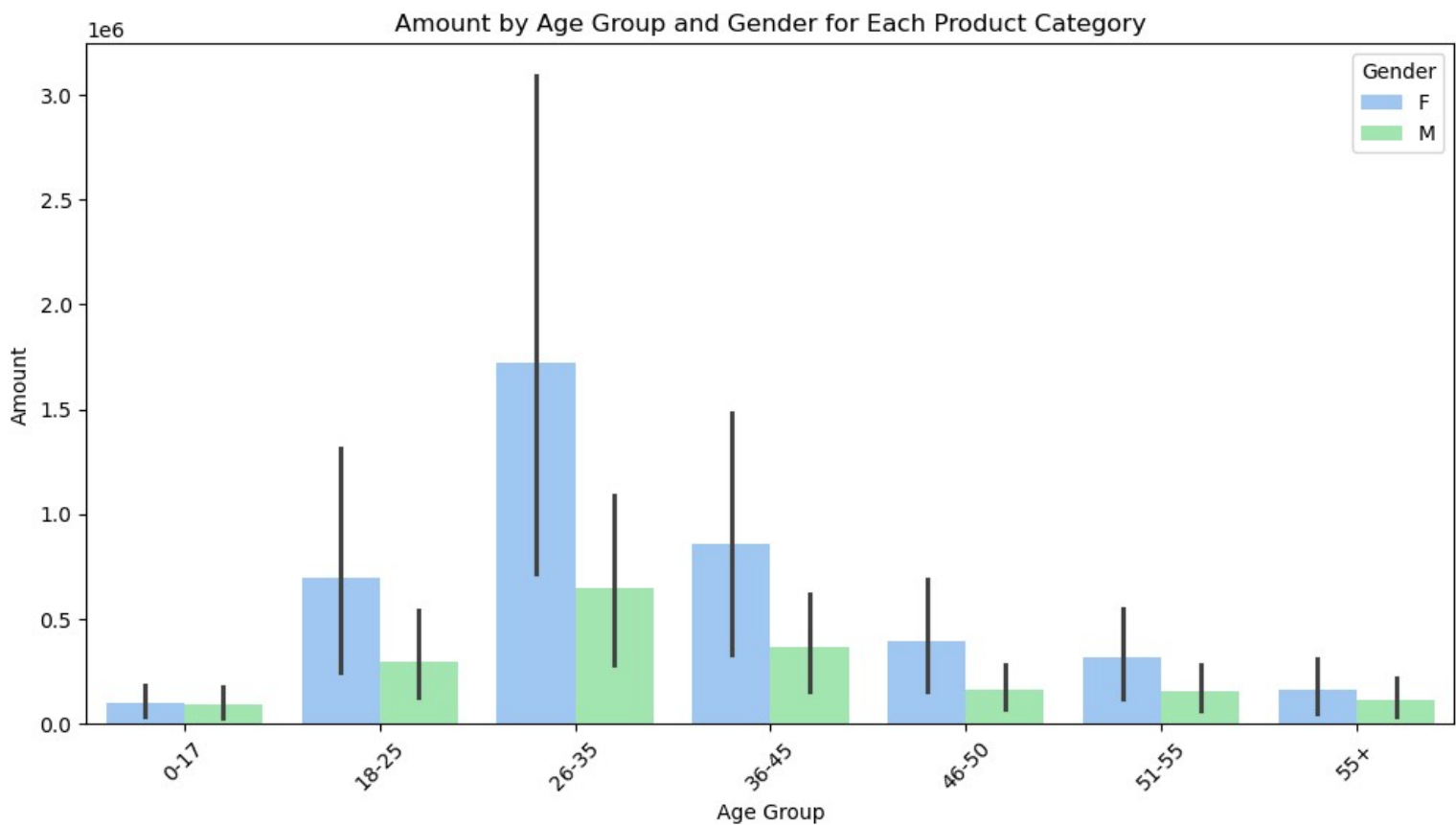
Further Exploration

Which age group contributes the most to each product category, and does this vary by gender?

```
age_cat_amount = df.groupby(['Product_Category', 'Age Group', 'Gender'])
['Amount'].sum().reset_index()
age_cat_amount.head(10)
```

	Product_Category	Age Group	Gender	Amount
0	Auto	0-17	F	9732
1	Auto	0-17	M	47151
2	Auto	18-25	F	288925
3	Auto	18-25	M	61369
4	Auto	26-35	F	541529
5	Auto	26-35	M	353578
6	Auto	36-45	F	263399
7	Auto	36-45	M	61121
8	Auto	46-50	F	61132
9	Auto	46-50	M	18608

```
plt.figure(figsize=(12,6))
sns.barplot(data=age_cat_amount, x='Age Group', y='Amount', hue='Gender')
plt.title("Amount by Age Group and Gender for Each Product Category")
plt.xticks(rotation=45)
plt.show()
```

Key Observations

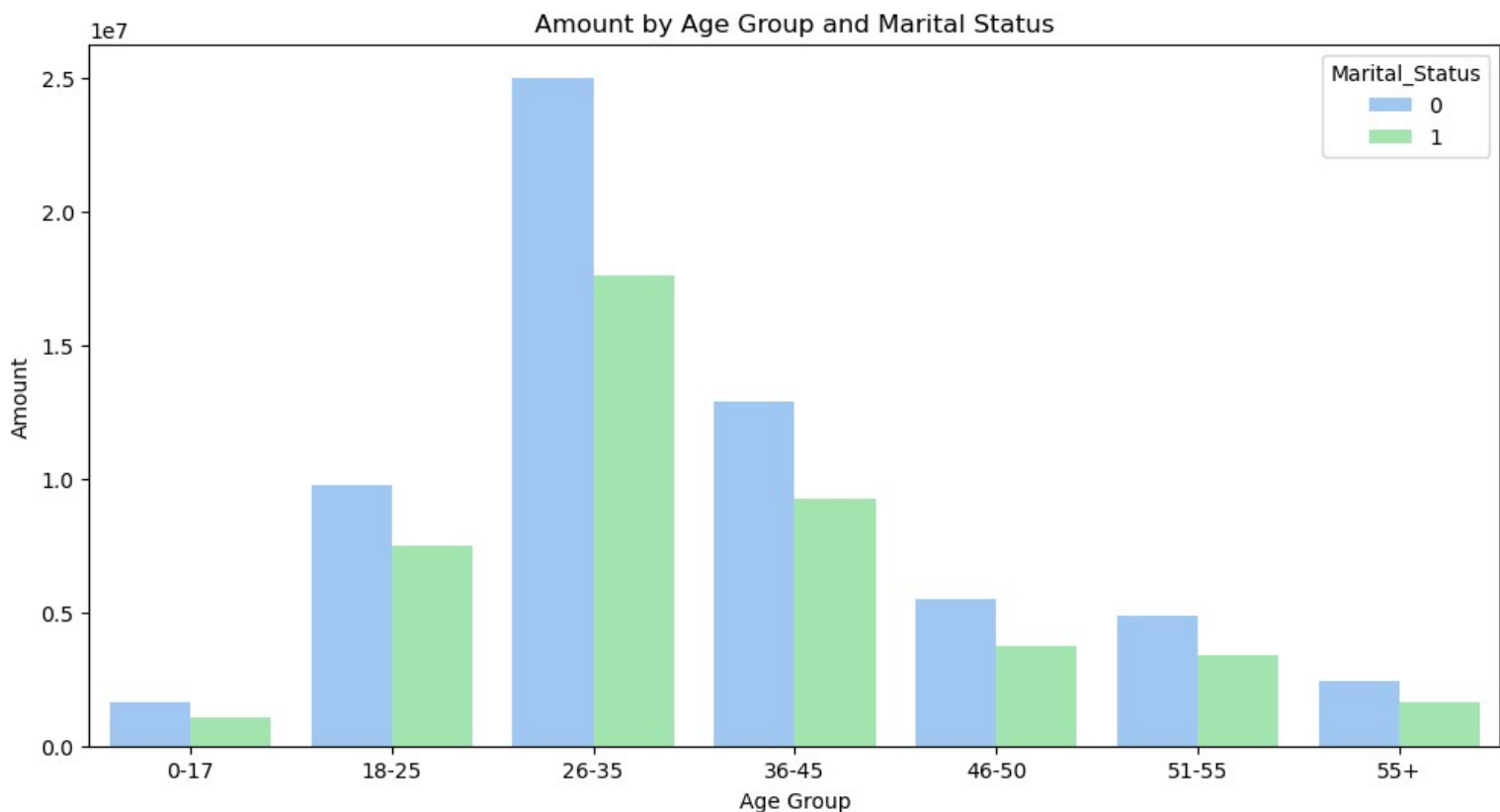
- 26–35 is the dominant age group for both genders, but female spend in this bracket (~ ₹1.72M) is nearly x3 that of males (~ ₹0.64M)
- Female dominance is consistent across every age group without exception — there is no age group where males outspend females
- The gender gap narrows in older age groups (51–55, 55+), suggesting more balanced spending behaviour post-50
- 18–25 females are a notable secondary segment, spending more than males aged 26–35

How does the amount spent vary by marital status across different age groups?

```
marital_age_amount = df.groupby(['Marital_Status', 'Age Group'])
['Amount'].sum().reset_index()
marital_age_amount.head()
```

	Marital_Status	Age Group	Amount
0	0	0-17	1655299
1	0	18-25	9781617
2	0	26-35	25008749
3	0	36-45	12870401
4	0	46-50	5472555

```
plt.figure(figsize=(12,6))
sns.barplot(data=marital_age_amount, x='Age Group', y='Amount', hue='Marital_Status')
plt.title("Amount by Age Group and Marital Status")
plt.show()
```



Key Observations

- Unmarried customers outspend married customers in every single age group — the pattern is universal and consistent
- The 26–35 unmarried segment is the absolute peak, spending ~₹2.50 crore — nearly 43% more than their married counterparts in the same age group
- Both marital groups follow the same inverted-V spending curve, peaking at 26–35 and declining steadily after 45
- The gap between unmarried and married narrows significantly in older age groups (46+), suggesting marital status becomes less influential as customers age
- 18–25 unmarried is a notable emerging segment — their spend (~₹0.98 crore) rivals married 36–45 customers

Which states show the highest growth in orders and revenue, and are there seasonal spikes in sales?

```
# Create a synthetic 'Order_Date' column
# This assumes orders are spread daily over a period starting from '2025-01-01'
df['Order_Date'] = pd.to_datetime(pd.date_range(start='2025-01-01', periods=len(df),
freq='D')).to_series().reset_index(drop=True))

# Create Month-Year column
df['Month_Year'] = df['Order_Date'].dt.to_period('M').astype(str)

df['Month_Year'] = pd.Categorical(pd.cut(np.arange(len(df)), bins=12, labels=[f'2025-
{str(m).zfill(2)}' for m in range(1,13)]))

# Group by State and Month_Year to get orders and revenue
state_month = df.groupby(['State', 'Month_Year'], observed=True).agg(Orders=('Orders',
'sum'),Revenue=('Amount', 'sum')).reset_index()

state_month.head()
```

	State	Month_Year	Orders	Revenue
0	Andhra Pradesh	2025-01	228	1784375
1	Andhra Pradesh	2025-02	149	968343
2	Andhra Pradesh	2025-03	117	728590
3	Andhra Pradesh	2025-04	132	639167
4	Andhra Pradesh	2025-05	244	1006059

```

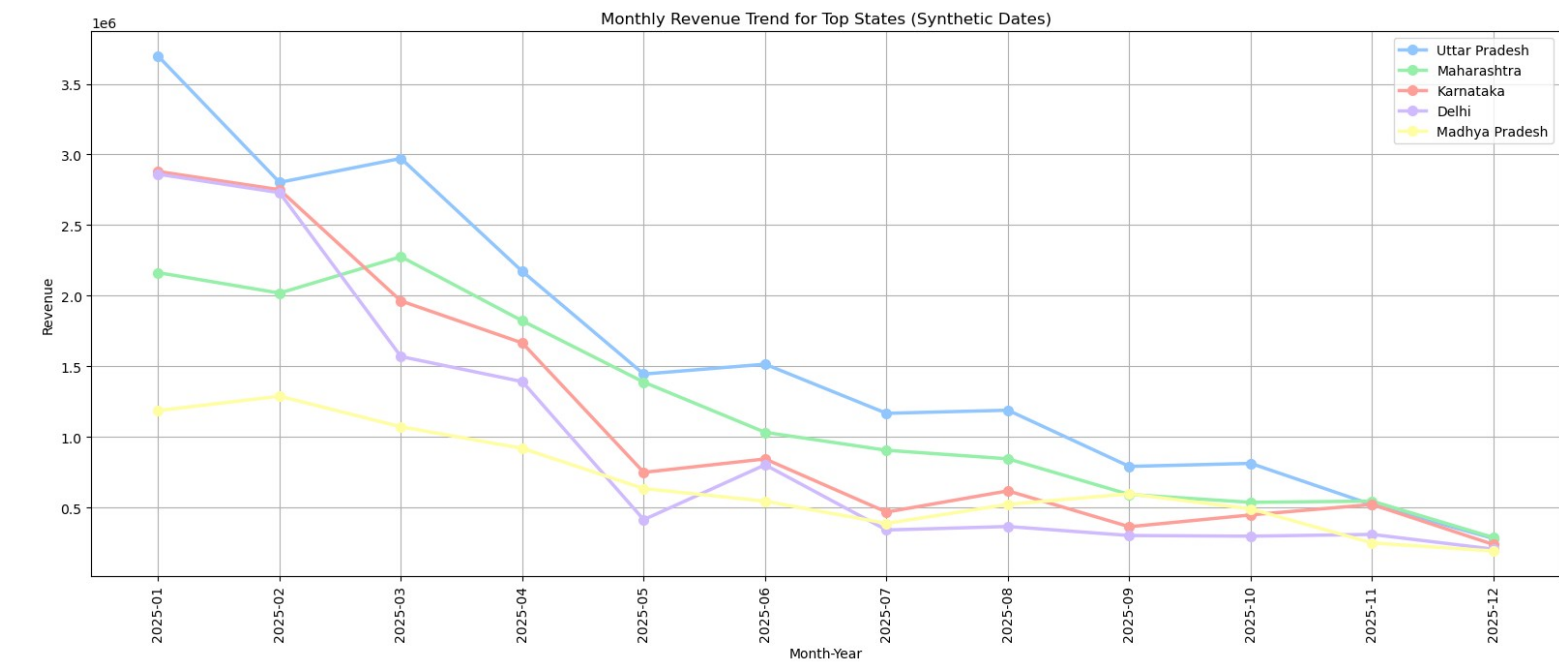
top_state_amount = df.groupby('State')['Amount'].sum().nlargest(5).reset_index()
top_state_amount.columns = ['State', 'Revenue']

top_states = top_state_amount['State'].head(5).tolist()# from earlier

plt.figure(figsize=(16, 7))
for i, st in enumerate(top_states):
    temp = state_month[state_month['State'] == st].sort_values('Month_Year')
    plt.plot(temp['Month_Year'], temp['Revenue'],
             marker='o',
             label=st,
             linewidth=2.5,
             markersize=7)

plt.xticks(rotation=90)
plt.title("Monthly Revenue Trend for Top States (Synthetic Dates)")
plt.xlabel("Month-Year")
plt.ylabel("Revenue")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```



This dataset does not contain transaction dates. Monthly revenue trends shown use synthetic dates for illustrative purposes only and do not reflect real seasonality. All other analyses (demographics, geography, product categories) are based on actual data.

Are there specific occupations that prefer particular product categories more than others?

```
occ_cat = df.groupby(['Occupation', 'Product_Category'])['Amount'].sum().reset_index()

pivot_occ_cat = occ_cat.pivot(index='Occupation', columns='Product_Category',
                                values='Amount').fillna(0)
pivot_occ_cat
```

Product_Category	Auto	Beauty	Books	Clothing & Apparel	Decor \
Occupation					
Agriculture	18609.0	43951.0	28650.0	437296.0	14057.0
Automobile	165586.0	94231.0	10364.0	870150.0	27818.0
Aviation	244129.0	184171.0	222422.0	1716254.0	59625.0
Banking	253553.0	153034.0	109905.0	1738400.0	128764.0
Chemical	0.0	53654.0	73531.0	727491.0	29494.0
Construction	80219.0	72604.0	29007.0	678794.0	36853.0
Food Processing	146326.0	45802.0	28568.0	603436.0	41789.0
Govt	245267.0	183985.0	85864.0	1158860.0	19943.0
Healthcare	103510.0	289418.0	97037.0	2249961.0	101273.0
Hospitality	150904.0	93428.0	18561.0	1131531.0	59467.0
IT Sector	211895.0	335291.0	166475.0	2226012.0	115576.0
Lawyer	71184.0	80797.0	83539.0	771556.0	6111.0
Media	163427.0	137187.0	39347.0	895328.0	41830.0
Retail	84720.0	134617.0	44908.0	744577.0	17834.0
Textile	19280.0	57314.0	23300.0	545373.0	29926.0

Product_Category	Electronics & Gadgets	Food	Footwear & Shoes \
Occupation			
Agriculture	352614.0	930908.0	336636.0
Automobile	1006127.0	1446125.0	717888.0
Aviation	1774366.0	4306343.0	1972899.0
Banking	1919127.0	2894943.0	1611992.0
Chemical	775674.0	1821161.0	864974.0
Construction	612017.0	860256.0	461129.0
Food Processing	796316.0	1047875.0	729173.0
Govt	972346.0	3242839.0	1068102.0
Healthcare	1802020.0	4465342.0	1820931.0
Hospitality	1098596.0	1727302.0	932455.0
IT Sector	2220174.0	4655906.0	2284568.0
Lawyer	508597.0	1780666.0	835034.0
Media	750193.0	2351254.0	728628.0
Retail	457335.0	1540485.0	726085.0
Textile	598344.0	862478.0	484715.0

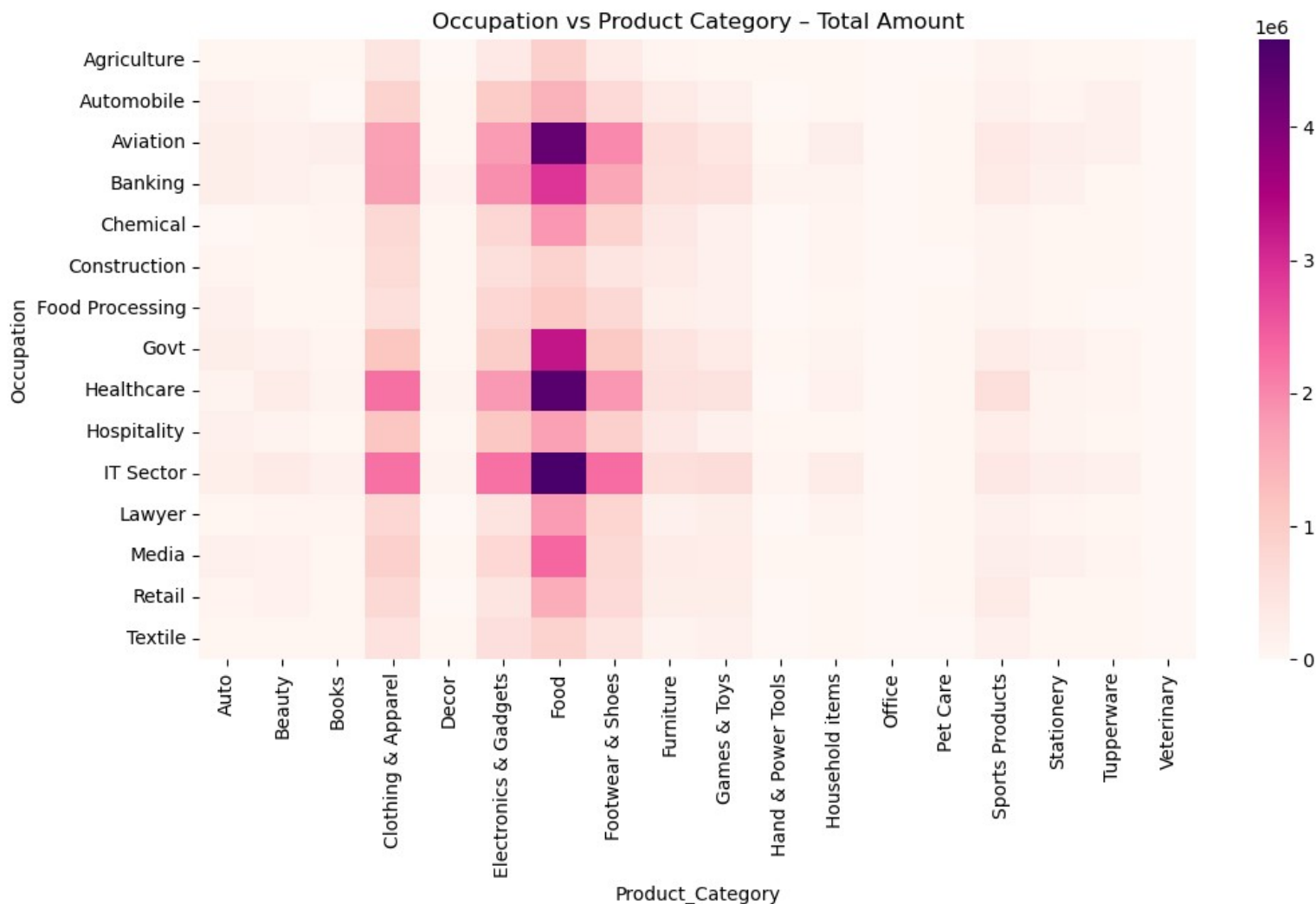
Product_Category	Furniture	Games & Toys	Hand & Power Tools \
Occupation			
Agriculture	80594.0	59005.0	23160.0
Automobile	339543.0	149516.0	0.0
Aviation	618951.0	429440.0	19324.0
Banking	605507.0	536829.0	123721.0
Chemical	400051.0	195728.0	0.0
Construction	299389.0	147294.0	15151.0
Food Processing	208654.0	189455.0	0.0
Govt	481088.0	337338.0	38536.0
Healthcare	563319.0	513805.0	14627.0
Hospitality	395549.0	193569.0	33937.0
IT Sector	616630.0	646992.0	85245.0

Lawyer	187075.0	252884.0	0.0
Media	277065.0	265881.0	37254.0
Retail	240251.0	249604.0	14663.0
Textile	126385.0	164354.0	0.0

Product_Category	Household items	Office	Pet Care	Sports Products \
Occupation				
Agriculture	44079.0	4175.0	14926.0	91762.0
Automobile	72656.0	931.0	26798.0	192694.0
Aviation	226272.0	7537.0	61840.0	360274.0
Banking	125833.0	6415.0	34061.0	307393.0
Chemical	76917.0	5514.0	23289.0	102553.0
Construction	74165.0	3341.0	13567.0	97884.0
Food Processing	51764.0	773.0	21384.0	99376.0
Govt	117981.0	5499.0	36174.0	276709.0
Healthcare	141446.0	11859.0	61895.0	603478.0
Hospitality	67468.0	3413.0	30925.0	255115.0
IT Sector	287214.0	16288.0	58731.0	387421.0
Lawyer	121994.0	5335.0	20414.0	147023.0
Media	54370.0	6065.0	38994.0	226115.0
Retail	52495.0	3054.0	24441.0	334112.0
Textile	54683.0	1737.0	14838.0	154024.0

Product_Category	Stationery	Tupperware	Veterinary
Occupation			
Agriculture	64976.0	42085.0	5604.0
Automobile	73226.0	162058.0	12885.0
Aviation	221390.0	166601.0	10460.0
Banking	170462.0	37405.0	13266.0
Chemical	72463.0	70683.0	4259.0
Construction	68010.0	37659.0	10172.0
Food Processing	37712.0	16311.0	5956.0
Govt	152984.0	87400.0	6297.0
Healthcare	107586.0	78600.0	8479.0
Hospitality	102354.0	71217.0	10614.0
IT Sector	226817.0	196154.0	17690.0
Lawyer	81314.0	25338.0	2804.0
Media	193386.0	88770.0	738.0
Retail	64829.0	45682.0	3478.0
Textile	38542.0	29679.0	0.0

```
plt.figure(figsize=(12,6))
sns.heatmap(pivot_occ_cat, annot=False, cmap= 'RdPu')
plt.title("Occupation vs Product Category – Total Amount")
plt.show()
```



Key Observations

Food dominates across all occupations

- The entire Food column is noticeably darker than any other — it's the universal top category regardless of profession
- IT, Healthcare, and Aviation show the darkest Food cells, consistent with their higher overall spend

Electronics & Gadgets is occupation-specific

- Strongest in IT Sector, Healthcare, and Aviation — high-income, tech-comfortable professions
- Nearly invisible for Agriculture, Textile, Construction — lower digital affinity segments

Clothing & Apparel shows broad but moderate appeal

- Visible across most occupations with light-to-medium shading
- No single occupation dominates this category

The right half of the heatmap is almost entirely blank

- Categories like Books, Decor, Office, Pet Care, Tupperware, Veterinary generate near-zero spend across all occupations
- These are either niche or not relevant to New Year shopping behavior

Agriculture, Textile, Construction are uniformly light

- Consistent with their low overall revenue contribution
- No strong category preference visible — low spend spread thinly

Strategic Insight:

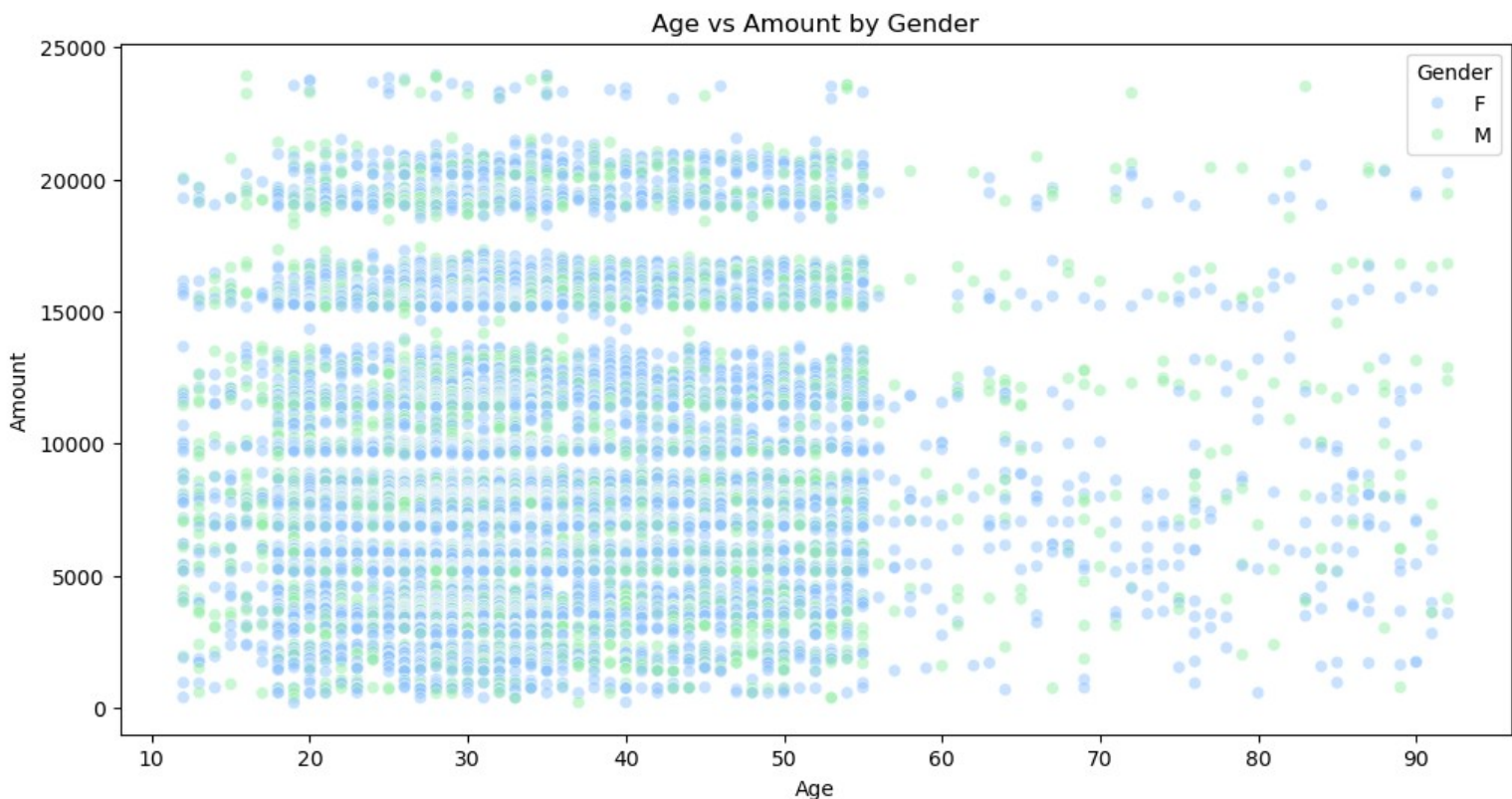
- Since Food is universally dominant, it should anchor all campaigns regardless of occupation targeting
- For IT/Healthcare/Aviation specifically, bundling Food + Electronics + Footwear offers would capture their full spending pattern
- Lower-income occupations (Agriculture, Textile) need value-focused food promotions rather than lifestyle or tech upsells.

What is the correlation between age and spending amount, and does this differ by gender?

```
df[['Age', 'Amount']].corr()
```

	Age	Amount
Age	1.000000	0.030941
Amount	0.030941	1.000000

```
plt.figure(figsize=(12,6))
sns.scatterplot(data=df, x='Age', y='Amount', hue='Gender', alpha=0.5)
plt.title("Age vs Amount by Gender")
plt.show()
```



Key Observations

- Correlation = 0.031 (Age alone cannot predict how much someone spends)
- Uniform spend across ages (All age groups have both high and low spenders)
- No gender-age interaction (Gender and age don't combine to create spending trends)
- Dense 15–50 cluster (This is where volume comes from, not higher per-order spend)

Key Insight:

- Spending amount in this dataset is driven by what people buy (product category) and who they are (occupation, marital status) — not how old they are
- Age groups differ in total revenue only because of differences in customer count, not because older or younger customers spend more per transaction.